

Dv Mini Project Report

Name - Swanit Nitin Bhawar

PRN - 202501110203

Branch – CSE(AIML)

Class – AL2,A25

Subject - Data Visualization

Project Title - CHOCOLATE SALES ANALYTICS DASHBOARD

Dataset Chocolate Sales Analytics - <https://github.com/chandoo-org/Power-BI>

Powerbi Published Link - https://app.powerbi.com/links/33HpTPDogy?ctid=7e394fc8-4b86-4cfe-810e-43f86f8bec47&pbi_source=linkShare

Explanation Video -

https://drive.google.com/file/d/1usFVPioqhZdOYPnmNbJvVejl7WG-_3re/view?usp=sharing

ABSTRACT :

The Chocolate Sales Analytics Dashboard is a comprehensive Power BI project developed to analyze sales performance, customer behavior, product contribution, shipment distribution, and sustainability impact using interactive visualizations. The dashboard enables business users to monitor KPIs, identify trends, compare performance over time, and gain actionable insights through multiple dashboard pages.

The project utilizes advanced Power BI concepts such as data modeling, DAX measures, calculated columns, time intelligence functions, bookmarks, slicers, and interactive navigation. The dashboard also integrates Sustainable Development Goal (SDG) 12 to demonstrate sustainability analysis alongside business intelligence

OBJECTIVES

The main objectives of the project are:

- To analyze overall sales and profit performance.
- To visualize product-wise and regional sales contribution.
- To implement interactive dashboards using bookmarks and slicers.
- To perform customer and shipment analysis.
- To apply time intelligence functions for trend analysis.
- To integrate SDG Goal 12 sustainability insights.
- To create a professional business intelligence solution using Power BI.

DATASET DESCRIPTION -

The dataset used in this project contains chocolate sales-related information including:

- Sales data
- Product categories
- Shipment details
- Customer information
- Regional sales data
- Time-based sales records

The dataset consists of multiple related tables connected using a star schema model.

1.Data Loading –

Navigator

chocolate dataset.xlsx [8]

calendar

locations

people

products

shipments

Calendar Table

Dimension Data

Shipment Data

shipments

Preview downloaded on Saturday, May 2, 2026

Sales Person	Geography	Product	Date	Sales	Boxes
Husein Augar	USA	Raspberry Choco	10-Mar-23	6657.75	222
Curtice Advani	New Zealand	85% Dark Bars	19-Oct-23	5202	454
Mallorie Waber	UK	Raspberry Choco	02-Feb-23	1545.75	213
Rafaelita Blaksland	UK	Raspberry Choco	22-Aug-23	1496.25	211
Madelene Upcott	Australia	Organic Choco Syrup	22-Jan-24	1383.75	369
Kelci Walkden	UK	Caramel Stuffed Bars	04-Apr-23	711	930
Oby Sorrel	India	Manuka Honey Choco	23-Nov-23	3327.75	142
Jehu Rudeforth	New Zealand	Milk Bars	30-Nov-23	2535.75	151
Ches Bonnell	New Zealand	White Choc	08-Aug-23	6943.5	58
Camilla Castle	UK	Almond Choco	20-Mar-23	13776.75	106
Barr Faughny	Australia	Eclairs	28-Jun-23	5580	295
Husein Augar	USA	Drinking Coco	22-Mar-23	4925.25	527
Oby Sorrel	New Zealand	50% Dark Bites	16-Mar-23	5384.25	242
Karlen McCaffrey	UK	Choco Coated Almonds	01-Jan-24	3215.25	496
Barr Faughny	India	Manuka Honey Choco	29-Mar-23	4641.75	12
Oby Sorrel	New Zealand	Baker's Choco Chips	21-Feb-24	5789.25	56
Ches Bonnell	Canada	Almond Choco	12-Jul-23	1147.5	30
Camilla Castle	USA	Choco Coated Almonds	02-Feb-23	5496.75	65
Wilone O'Kielt	Australia	Caramel Stuffed Bars	06-Feb-24	717.75	527
Ches Bonnell	New Zealand	Manuka Honey Choco	17-May-23	157.5	419
Dotty Strutley	Canada	50% Dark Bites	19-Oct-23	10539	837
Ian Morforth	Australia	Eclairs	07-Jul-23	4918.5	207

Load

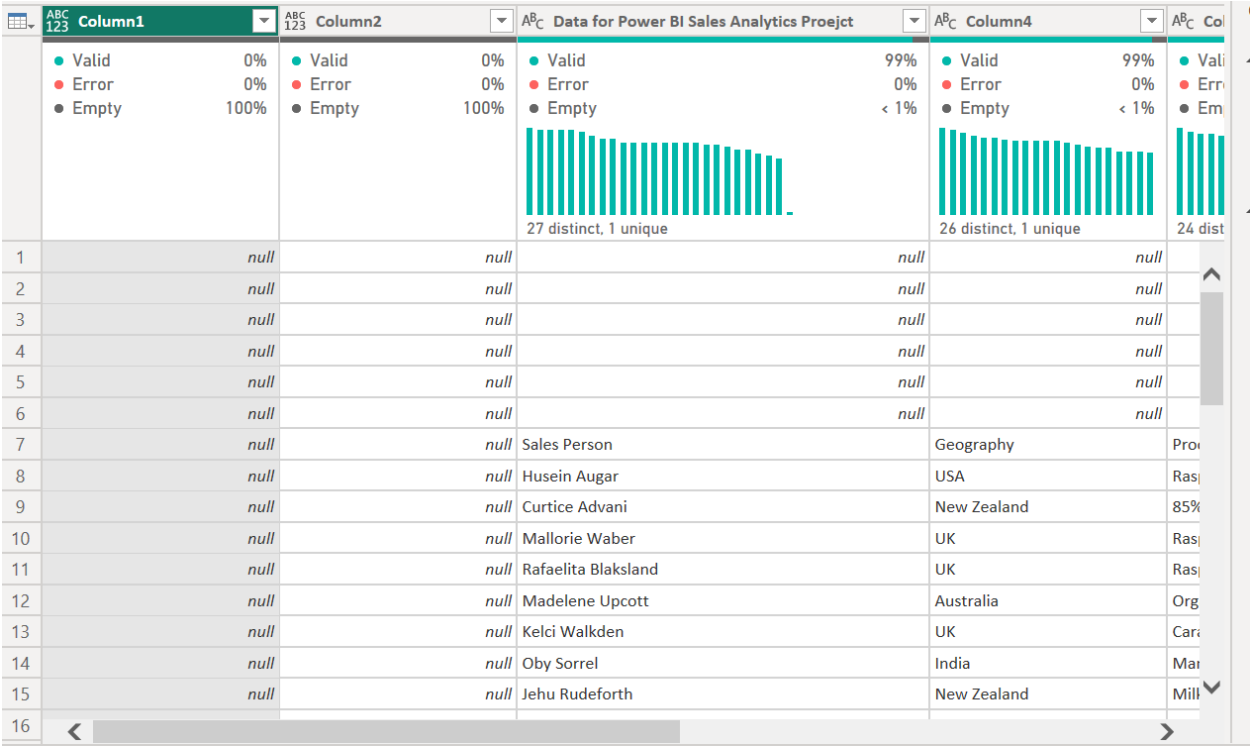
Transform Data

Cancel

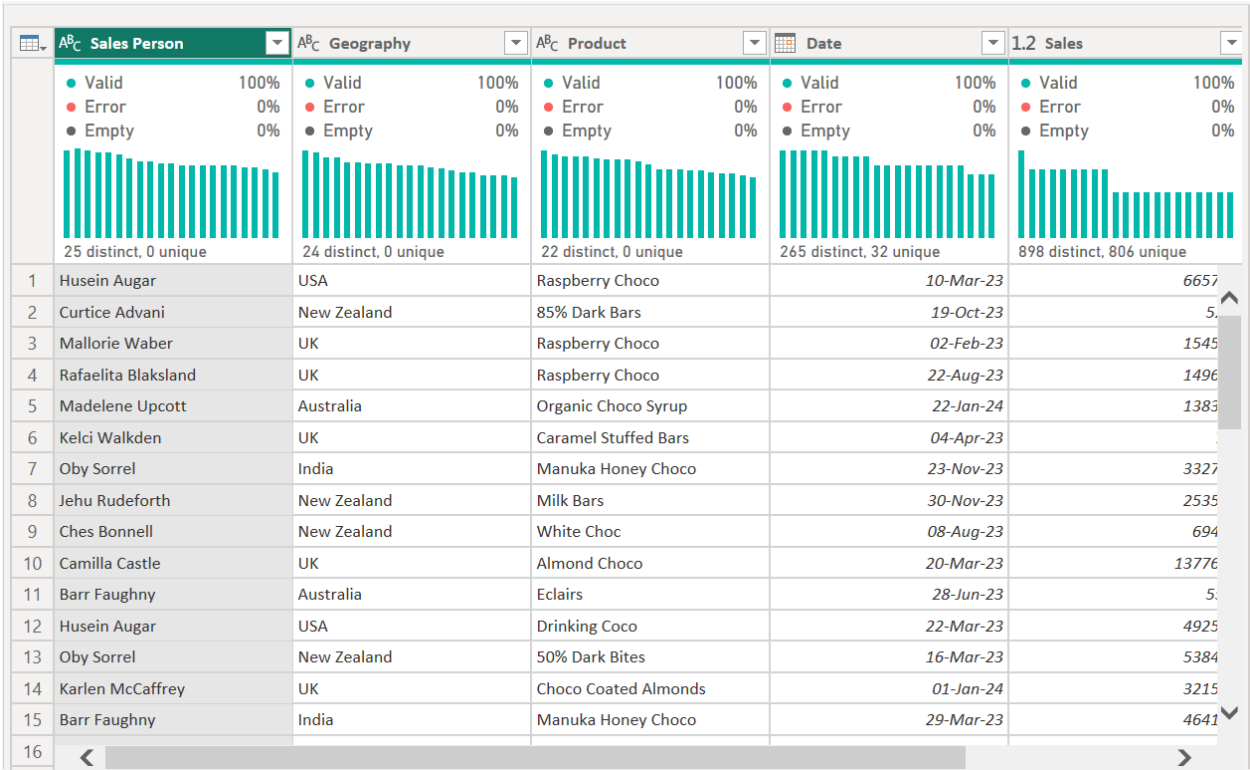
2.Data Transformation –

Null Values are removed

Before



After



3. Relationship Model – The project follows a Star Schema data model where the Shipments table acts as the central fact table connected to multiple dimension tables.

Fact Table

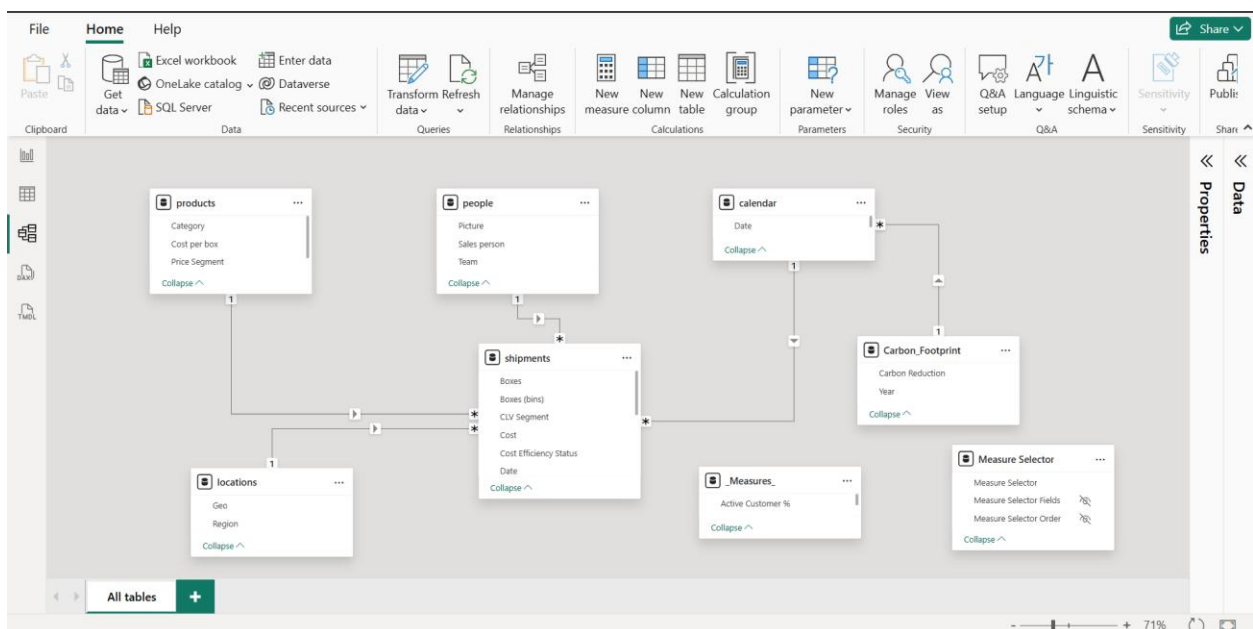
- Shipments

Dimension Tables

- Products
- People
- Locations
- Calendar

The relationships were created using one-to-many relationships between dimension tables and the fact table.

Star Schema is used



File Home Help
Table tools
Share

Name

Manage
relationships
Relationships

New
measure
Calculations

Quick
measure
Calculations

New
column
Calculations

New
table
Calculations

Mark as date
table
Calendars

Date	Year	Month	Month Name	Day of Week	Day Name	Start of Month	Month Number
Wednesday, February 1, 2023	2023	2	February		3 Wednesday	Wednesday, February 1, 2023	2
Thursday, February 2, 2023	2023	2	February		4 Thursday	Wednesday, February 1, 2023	2
Friday, February 3, 2023	2023	2	February		5 Friday	Wednesday, February 1, 2023	2
Saturday, February 4, 2023	2023	2	February		6 Saturday	Wednesday, February 1, 2023	2
Sunday, February 5, 2023	2023	2	February		0 Sunday	Wednesday, February 1, 2023	2
Monday, February 6, 2023	2023	2	February		1 Monday	Wednesday, February 1, 2023	2
Tuesday, February 7, 2023	2023	2	February		2 Tuesday	Wednesday, February 1, 2023	2
Wednesday, February 8, 2023	2023	2	February		3 Wednesday	Wednesday, February 1, 2023	2
Thursday, February 9, 2023	2023	2	February		4 Thursday	Wednesday, February 1, 2023	2
Friday, February 10, 2023	2023	2	February		5 Friday	Wednesday, February 1, 2023	2
Saturday, February 11, 2023	2023	2	February		6 Saturday	Wednesday, February 1, 2023	2
Sunday, February 12, 2023	2023	2	February		0 Sunday	Wednesday, February 1, 2023	2
Monday, February 13, 2023	2023	2	February		1 Monday	Wednesday, February 1, 2023	2
Tuesday, February 14, 2023	2023	2	February		2 Tuesday	Wednesday, February 1, 2023	2
Wednesday, February 15, 2023	2023	2	February		3 Wednesday	Wednesday, February 1, 2023	2
Thursday, February 16, 2023	2023	2	February		4 Thursday	Wednesday, February 1, 2023	2
Friday, February 17, 2023	2023	2	February		5 Friday	Wednesday, February 1, 2023	2
Saturday, February 18, 2023	2023	2	February		6 Saturday	Wednesday, February 1, 2023	2
Sunday, February 19, 2023	2023	2	February		0 Sunday	Wednesday, February 1, 2023	2
Monday, February 20, 2023	2023	2	February		1 Monday	Wednesday, February 1, 2023	2
Tuesday, February 21, 2023	2023	2	February		2 Tuesday	Wednesday, February 1, 2023	2

Table: calendar (394 rows)

Data

- > Measures_
- > calendar
- > Date
 - Day Name
 - Σ Day of Week
 - Σ Month
 - Month Name
 - Month Number
 - Start of Month
 - Year
- > Carbon Footprint
- > locations
- > Measure Selector
- > people
- > products
- > shipments

Dax used -

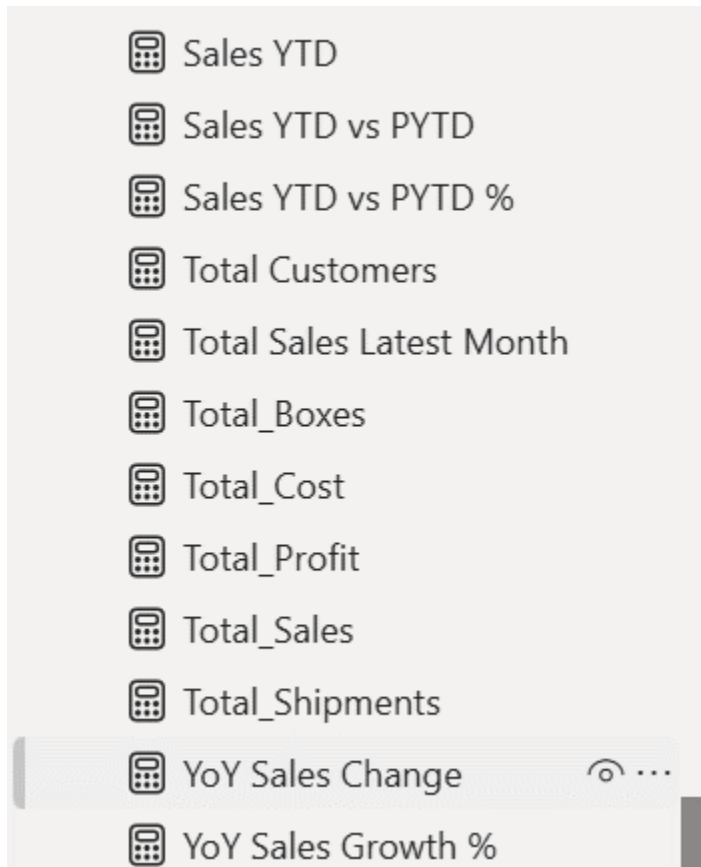
Table: calendar (394 rows) Column: Month Number (12 distinct values)

Date	Year	Month	Month Name	Day of Week	Day Name	Start of Month	Month Number
Wednesday, February 1, 2023	2023	2	February	3	Wednesday	Wednesday, February 1, 2023	2
Thursday, February 2, 2023	2023	2	February	4	Thursday	Wednesday, February 1, 2023	2
Friday, February 3, 2023	2023	2	February	5	Friday	Wednesday, February 1, 2023	2
Saturday, February 4, 2023	2023	2	February	6	Saturday	Wednesday, February 1, 2023	2
Sunday, February 5, 2023	2023	2	February	0	Sunday	Wednesday, February 1, 2023	2
Monday, February 6, 2023	2023	2	February	1	Monday	Wednesday, February 1, 2023	2
Tuesday, February 7, 2023	2023	2	February	2	Tuesday	Wednesday, February 1, 2023	2
Wednesday, February 8, 2023	2023	2	February	3	Wednesday	Wednesday, February 1, 2023	2
Thursday, February 9, 2023	2023	2	February	4	Thursday	Wednesday, February 1, 2023	2
Friday, February 10, 2023	2023	2	February	5	Friday	Wednesday, February 1, 2023	2
Saturday, February 11, 2023	2023	2	February	6	Saturday	Wednesday, February 1, 2023	2
Sunday, February 12, 2023	2023	2	February	0	Sunday	Wednesday, February 1, 2023	2
Monday, February 13, 2023	2023	2	February	1	Monday	Wednesday, February 1, 2023	2
Tuesday, February 14, 2023	2023	2	February	2	Tuesday	Wednesday, February 1, 2023	2
Wednesday, February 15, 2023	2023	2	February	3	Wednesday	Wednesday, February 1, 2023	2
Thursday, February 16, 2023	2023	2	February	4	Thursday	Wednesday, February 1, 2023	2
Friday, February 17, 2023	2023	2	February	5	Friday	Wednesday, February 1, 2023	2
Saturday, February 18, 2023	2023	2	February	6	Saturday	Wednesday, February 1, 2023	2
Sunday, February 19, 2023	2023	2	February	0	Sunday	Wednesday, February 1, 2023	2
Monday, February 20, 2023	2023	2	February	1	Monday	Wednesday, February 1, 2023	2
Tuesday, February 21, 2023	2023	2	February	2	Tuesday	Wednesday, February 1, 2023	2

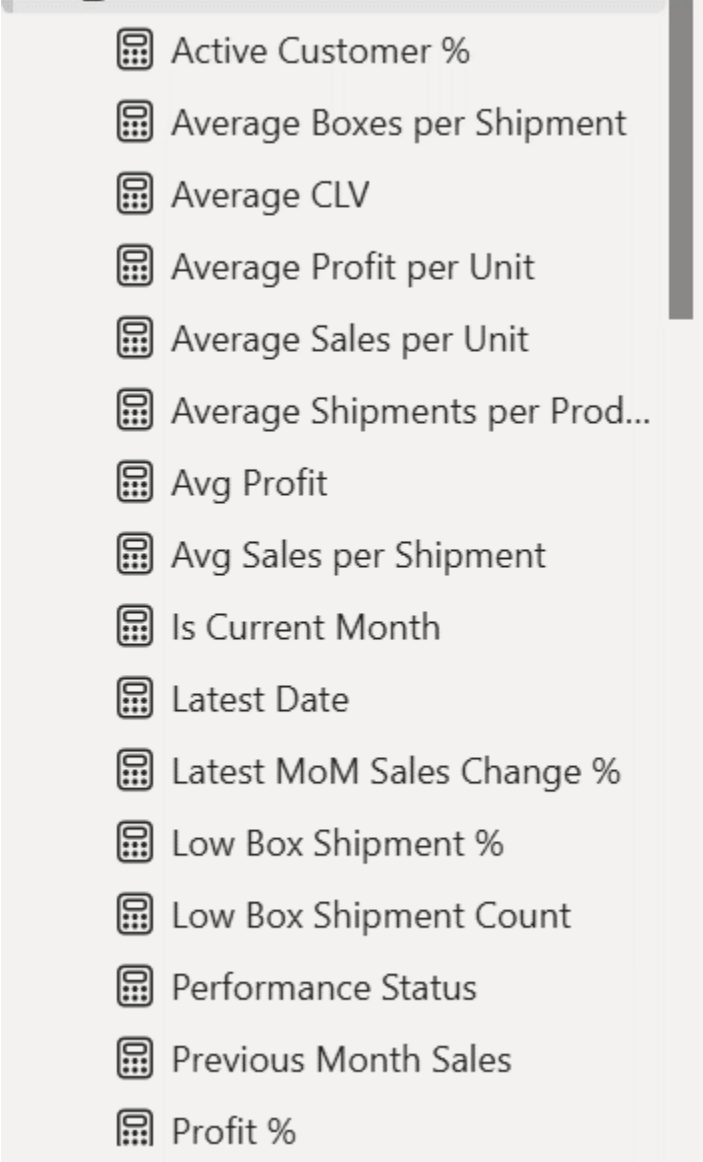
5.Measures-

- Total Sales
- Total Profit
- Total Shipments
- Total Boxes
- Profit %
- Sales Growth %
- Sales YoY
- Sales YTD
- Sales MTD
- Sales LY YTD
- Running Total Sales
- Repeat Purchase Rate %
- Average Sales per Unit
- Average Profit per Unit

- Average CLV



-  Profit Growth %
-  Repeat Purchase Rate %
-  Running Total Sales
-  Sales Growth %
-  Sales Last 30 Days
-  Sales Latest Year
-  Sales LY YTD  ...
-  Sales LY YTD Cumulative
-  Sales MoM
-  Sales MTD
-  Sales Performance vs PY %
-  Sales Previous Year
-  Sales PY
-  Sales PYTD
-  Sales SPLY
-  Sales YoY
-  Sales YoY %

- 
- Active Customer %
 - Average Boxes per Shipment
 - Average CLV
 - Average Profit per Unit
 - Average Sales per Unit
 - Average Shipments per Prod...
 - Avg Profit
 - Avg Sales per Shipment
 - Is Current Month
 - Latest Date
 - Latest MoM Sales Change %
 - Low Box Shipment %
 - Low Box Shipment Count
 - Performance Status
 - Previous Month Sales
 - Profit %

These measures help in analyzing business performance dynamically across multiple dashboard page

6. GRAIN OF DATA

The grain of the dataset represents shipment-level transactional data. Each row in the shipments table corresponds to an individual shipment transaction containing sales amount, boxes shipped, customer details, regional information, and shipment dates.

7. CALCULATED COLUMNS USED

1. Price Segment

Used to classify products into Premium, Mid Range, and Economy categories.

2. Year

Extracted year values from the calendar table for yearly analysis.

3. Month Name

Used for monthly trend visualization.

4. Quarter

Used for quarterly time analysis.

5. Shipment Size

Classified shipments into Large and Small categories based on shipment quantity.

8. TIME INTELLIGENCE FUNCTIONS USED

The dashboard uses advanced DAX time intelligence functions for trend analysis.

Time Intelligence Measures

- TOTALYTD()
- TOTALMTD()
- SAMEPERIODLASTYEAR()
- DATEADD()
- DATESYTD()
- DATESMTD()
- Running Total Calculations

These functions were used to calculate:

- Year-to-Date Sales
- Month-to-Date Sales
- Previous Year Sales
- YoY Growth
- Monthly Growth Trends
- Running Sales Totals

FileHomeHelpTable tools

Share

NameMeasure Selector

Manage relationships

New measure measure column

Quick

New table

New table

Mark as date table

Calendars

Structure

Relationships

Calculations

Calendars

1 Measure Selector = {

2 ("Boxes", NAMEOF('._Measures_.'[Total_Boxes]), 0),

3 ("Cost", NAMEOF('._Measures_.'[Total_Cost]), 1),

4 ("Profit", NAMEOF('._Measures_.'[Total_Profit]), 2),

5 ("Sales", NAMEOF('._Measures_.'[Total_Sales]), 3),

6 ("Shipments", NAMEOF('._Measures_.'[Total_Shipments]), 4)

7 }

Measure Selector

Measure Selector Fields

Measure Selector Order

Boxes

Cost

Profit

Sales

Shipments

._Measures_.[Total_Boxes]

._Measures_.[Total_Cost]

._Measures_.[Total_Profit]

._Measures_.[Total_Sales]

._Measures_.[Total_Shipments]

0

1

2

3

4

Data

Search

._Measures_

calendar

Carbon_Footprint

locations

Geo

Region

Measure Selector

Measure Selector

Measure Selector Fields

Measure Selector Order

people

products

shipments

Table: Measure Selector (5 rows)

FileHomeHelpTable toolsColumn tools

Share

NamePrice Segment

FormatText

SummarizationDon't summarize

Sort by column

Data groups

Manage relationships

New column

Calculations

Structure

Formatting

Properties

Sort

Groups

Relationships

Calculations

1 Price Segment =

2 IF(

3 products[Cost per box] >= 8,

4 "Premium",

5 IF(

6 products[Cost per box] >= 4,

7 "Mid Range",

8 "Economy"

9)

10)

Product

Category

Cost per box

Price Segment

Milk Bars

Bars

5.26

Mid Range

50% Dark Bites

Bites

7.48

Mid Range

Almond Choco

Bars

5.15

Mid Range

Raspberry Choco

Bars

3.85

Economy

Mint Chip Choco

Bars

5.72

Mid Range

Eclairs

Bites

6.31

Mid Range

Drinking Coco

Other

9.94

Premium

99% Dark & Pure

Bars

7.73

Mid Range

Orange Choco

Bars

3.68

Economy

Spicy Special Slims

Bites

8.22

Premium

After Nines

Bites

10.23

Premium

Fruit & Nut Bars

Bars

4.74

Mid Range

85% Dark Bars

Bars

10.51

Premium

Data

Search

._Measures_

calendar

Carbon_Footprint

locations

Geo

Region

Measure Selector

people

Picture

Sales person

Team

products

Category

Cost per box

Price Segment

Product

shipments

Table: products (22 rows) Column: Price Segment (3 distinct values)

FileHomeHelpTable toolsColumn tools

NameCLV Segment

FormatText

SummarizationDon't summarize

Sort by column

Data groups

Manage relationships

New column

Data typeText

Format\$ - %

Auto

Data categoryUncategorized

Structure

Formatting

Properties

Sort

Groups

Relationships

Calculations

1 CLV Segment =

2 IF(

3 shipments[Sales] >= 6000,

4 "High Value",

5 IF(

6 shipments[Sales] >= 3000,

7 "Medium Value",

8 "Low Value"

9)

10)

Profit Margin Category	Shipment Size Category	Sales Value Category	Cost Efficiency Status	Boxes (bins)	Sustainability Category	CLV Segment
High Profit	Small Shipment	Low Value	Efficient	0	Regular Line	Low Value
High Profit	Large Shipment	High Value	Efficient	800	Sustainable Line	High Value
High Profit	Large Shipment	High Value	Efficient	100	Regular Line	High Value
Low Profit	Medium Shipment	Low Value	High Cost	50	Regular Line	Low Value
High Profit	Large Shipment	Medium Value	Efficient	300	Sustainable Line	High Value
Low Profit	Large Shipment	Low Value	High Cost	600	Sustainable Line	Low Value
High Profit	Large Shipment	Low Value	Moderate	250	Regular Line	Low Value
High Profit	Large Shipment	Medium Value	Efficient	250	Regular Line	High Value
Low Profit	Large Shipment	Low Value	High Cost	100	Regular Line	Low Value
High Profit	Large Shipment	Medium Value	Efficient	150	Regular Line	High Value
High Profit	Medium Shipment	Medium Value	Efficient	50	Regular Line	Medium Value
High Profit	Small Shipment	Medium Value	Efficient	0	Regular Line	High Value

Table: shipments (6,113 rows) Column: CLV Segment (3 distinct values)

FileHomeHelpTable toolsColumn tools

NameCost Efficiency Status

FormatText

SummarizationDon't summarize

Sort by column

Data groups

Manage relationships

New column

Data typeText

Format\$ - %

Auto

Data categoryUncategorized

Structure

Formatting

Properties

Sort

Groups

Relationships

Calculations

1 Cost Efficiency Status =

2 VAR CostRatio = DIVIDE(shipments[Cost], shipments[Sales], 0)

3 RETURN

4 SWITCH(

5 TRUE(),

6 CostRatio <= 0.60, "Efficient",

7 CostRatio <= 0.80, "Moderate",

8 "High Cost"

9)

Cost	Profit	Profit Margin Category	Shipment Size Category	Sales Value Category	Cost Efficiency Status	Boxes (bins)	Sustainability Category
154.5	993	High Profit	Small Shipment	Low Value	Efficient	0	Regular Line
6260.76	4278.24	High Profit	Large Shipment	High Value	Efficient	800	Sustainable Line
1183.68	11400.57	High Profit	Large Shipment	High Value	Efficient	100	Regular Line
411.84	-229.59	Low Profit	Medium Shipment	Low Value	High Cost	50	Regular Line
1062.4	7548.35	High Profit	Large Shipment	Medium Value	Efficient	300	Sustainable Line
1694.64	-540.39	Low Profit	Large Shipment	Low Value	High Cost	600	Sustainable Line
1819.69	869.06	High Profit	Large Shipment	Low Value	Moderate	250	Regular Line
2118.02	4357.48	High Profit	Large Shipment	Medium Value	Efficient	250	Regular Line
1387.65	-962.4	Low Profit	Large Shipment	Low Value	High Cost	100	Regular Line
1331.44	5126.06	High Profit	Large Shipment	Medium Value	Efficient	150	Regular Line
739.8	5197.95	High Profit	Medium Shipment	Medium Value	Efficient	50	Regular Line
19.29	6663.21	High Profit	Small Shipment	Medium Value	Efficient	0	Regular Line
1194.8	6342.7	High Profit	Large Shipment	Medium Value	Efficient	200	Regular Line

Table: shipments (6,113 rows) Column: Cost Efficiency Status (3 distinct values)

9.DASHBOARD OVERVIEW

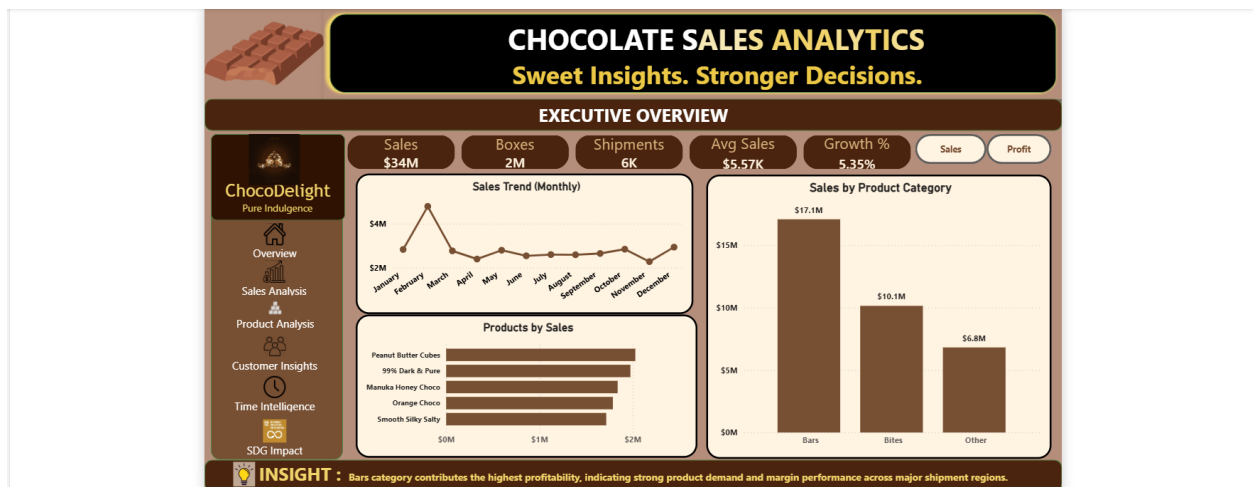
The project contains multiple interactive dashboard pages designed for business storytelling and analytics.

PAGE 1 – EXECUTIVE OVERVIEW

The Executive Overview dashboard provides a summarized business view of sales performance, shipment contribution, product category analysis, and monthly sales trends.

Features Included

- KPI Cards
- Monthly Sales Trend
- Product Sales Analysis
- Category Contribution
- Bookmark Navigation
- Interactive Storytelling

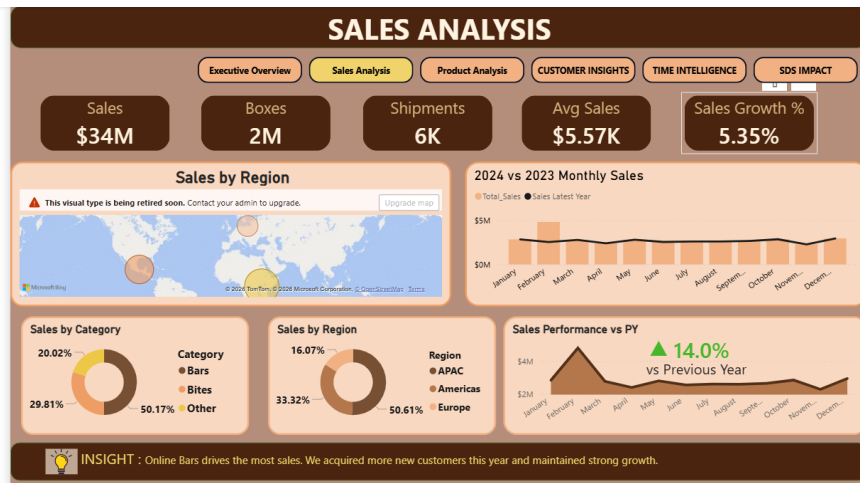


PAGE 2 – SALES ANALYSIS

The Sales Analysis dashboard focuses on regional performance and year-over-year sales comparison.

Features Included

- Sales by Region
- Monthly Sales Comparison
- Regional Contribution Analysis
- Sales Performance vs Previous Year
- Interactive Charts
- Insight Generation



PAGE 3 – PRODUCT ANALYSIS

The Product Analysis dashboard provides detailed insights into product contribution, shipment analysis, and price segment comparison.

Features Included

- Dynamic Bookmarks
- Sales / Profit / Boxes / Shipments Switching
- Product Category Analysis
- Price Segment Analysis
- Interactive Slicers
- Product Performance Visualization

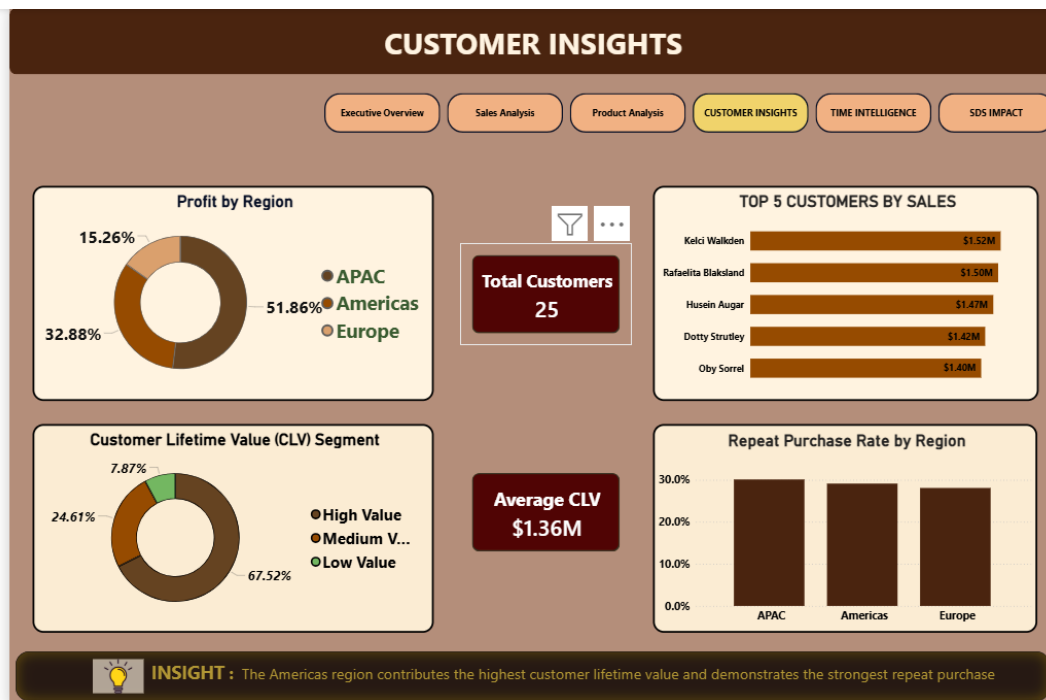


PAGE 4 – CUSTOMER INSIGHTS

The Customer Insights dashboard focuses on customer segmentation and repeat purchase behavior.

Features Included

- Customer Regional Distribution
- Top Customers Analysis
- Customer Lifetime Value (CLV)
- Repeat Purchase Rate
- Customer Segmentation

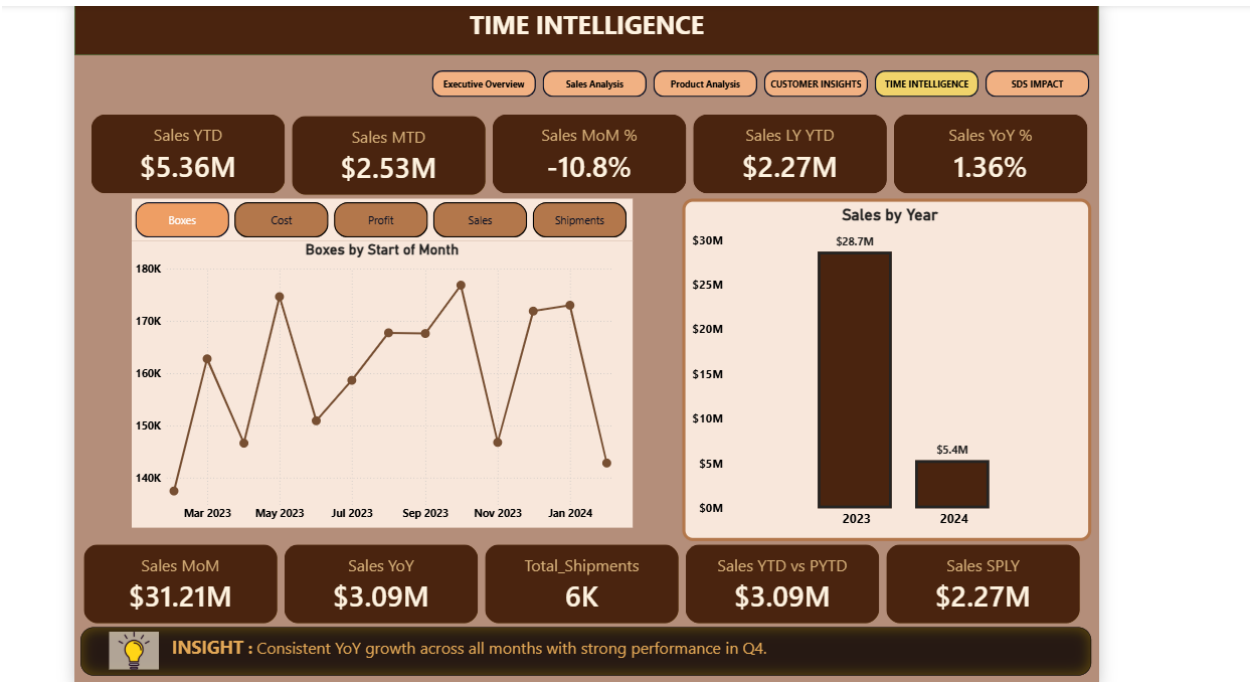


PAGE 5 – TIME INTELLIGENCE

The Time Intelligence dashboard provides time-based business analysis using advanced DAX functions.

Features Included

- Sales YTD
- Sales MTD
- Sales YoY
- Previous Year Comparison
- Running Total Sales
- Dynamic Metric Switching
- Time Trend Analysis

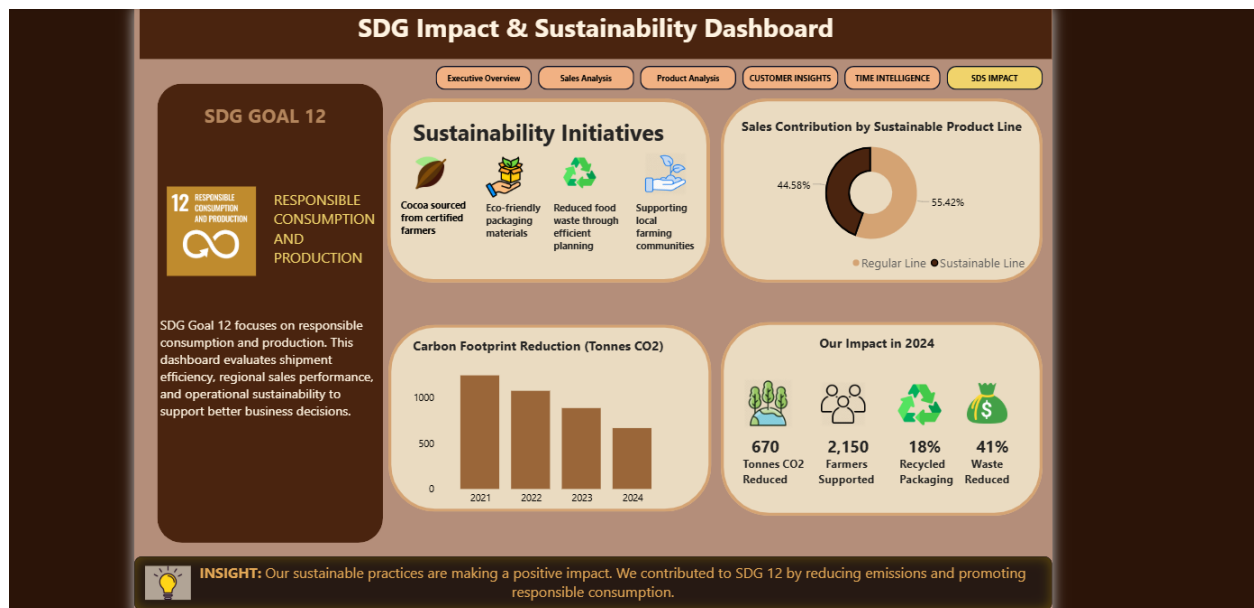


PAGE 6 – SDG IMPACT & SUSTAINABILITY

This dashboard maps the project to Sustainable Development Goal 12 – Responsible Consumption and Production.

Features Included

- Sustainability Initiatives
- Carbon Footprint Reduction
- Sustainable Product Contribution
- Environmental Impact Metrics
- SDG Goal Mapping



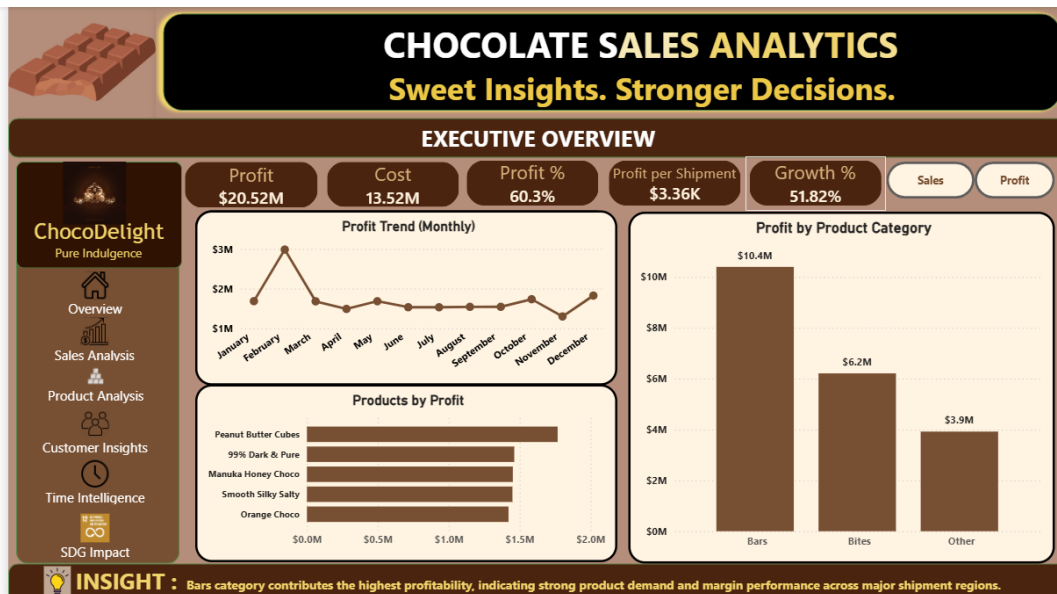
10. BOOKMARKS AND NAVIGATION

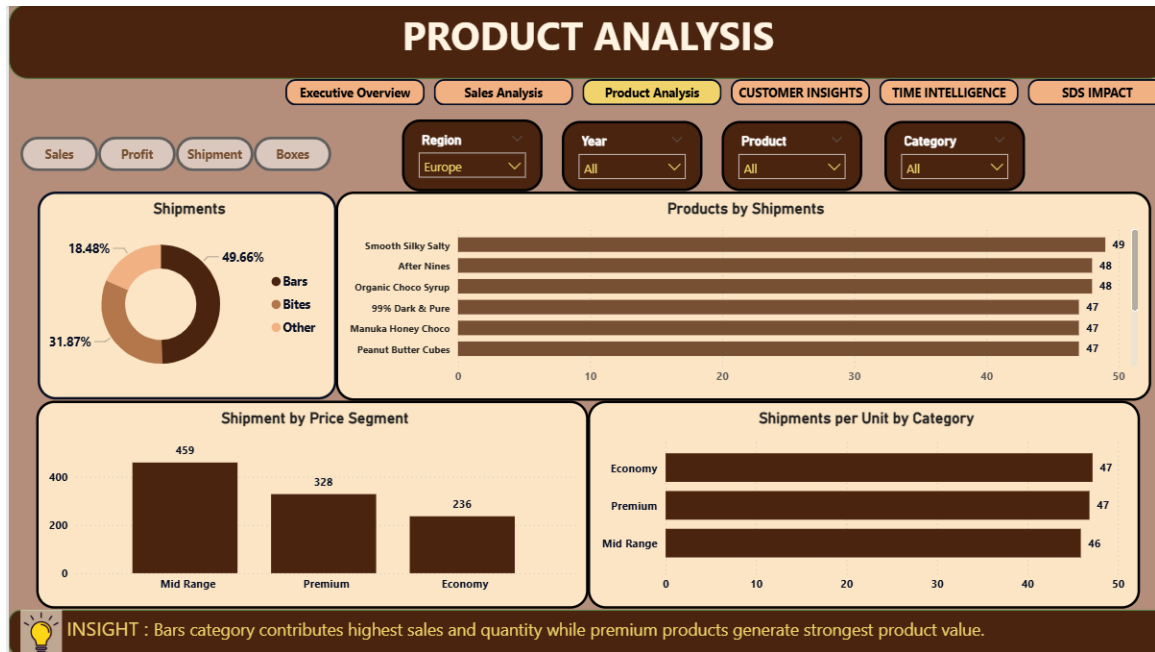
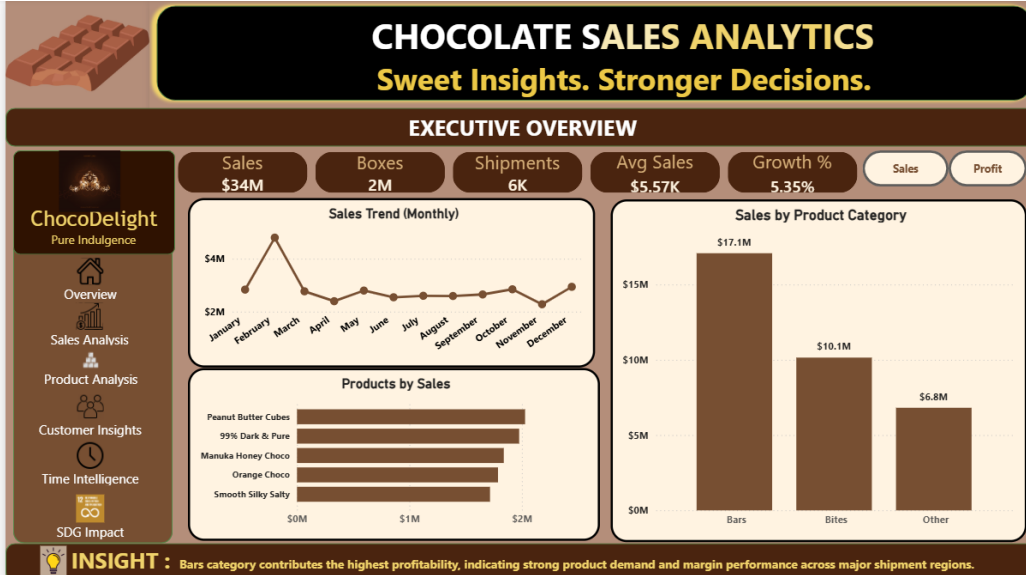
The dashboard uses bookmarks and navigation buttons to improve interactivity and user experience.

Bookmark Features

- Sales View
- Profit View
- Boxes View
- Shipments View

Bookmarks dynamically switch visuals and measures while maintaining the same dashboard layout.





PRODUCT ANALYSIS

Executive Overview

Sales Analysis

Product Analysis

CUSTOMER INSIGHTS

TIME INTELLIGENCE

SDS IMPACT

Sales

Profit

Shipment

Boxes

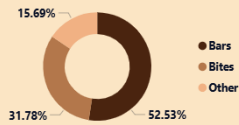
Region
Europe

Year
All

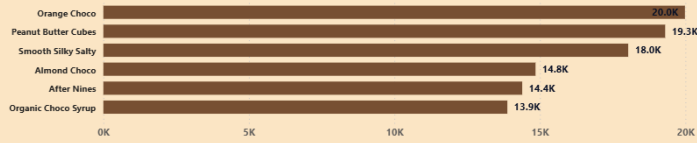
Product
All

Category
All

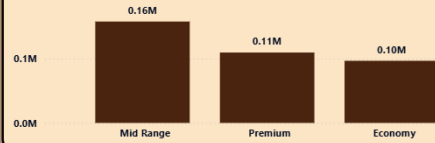
Boxes



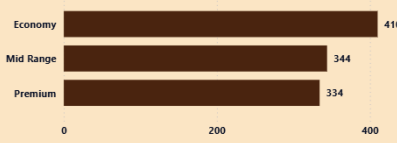
Products by Boxes



Boxes by Price Segment



Boxes per Unit by Category



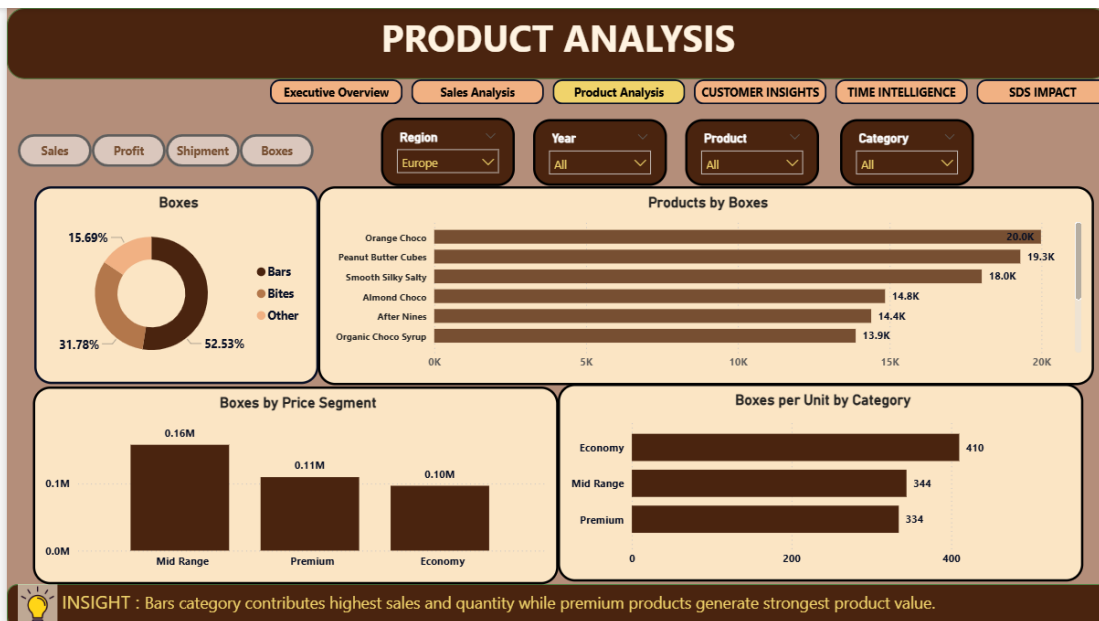
INSIGHT : Bars category contributes highest sales and quantity while premium products generate strongest product value.

11. INTERACTIVITY FEATURES

The dashboard includes:

- Slicers
- Filters
- Bookmarks
- Navigation Buttons
- Dynamic Visual Switching
- Interactive KPI Cards

These features improve dashboard usability and user engagement.



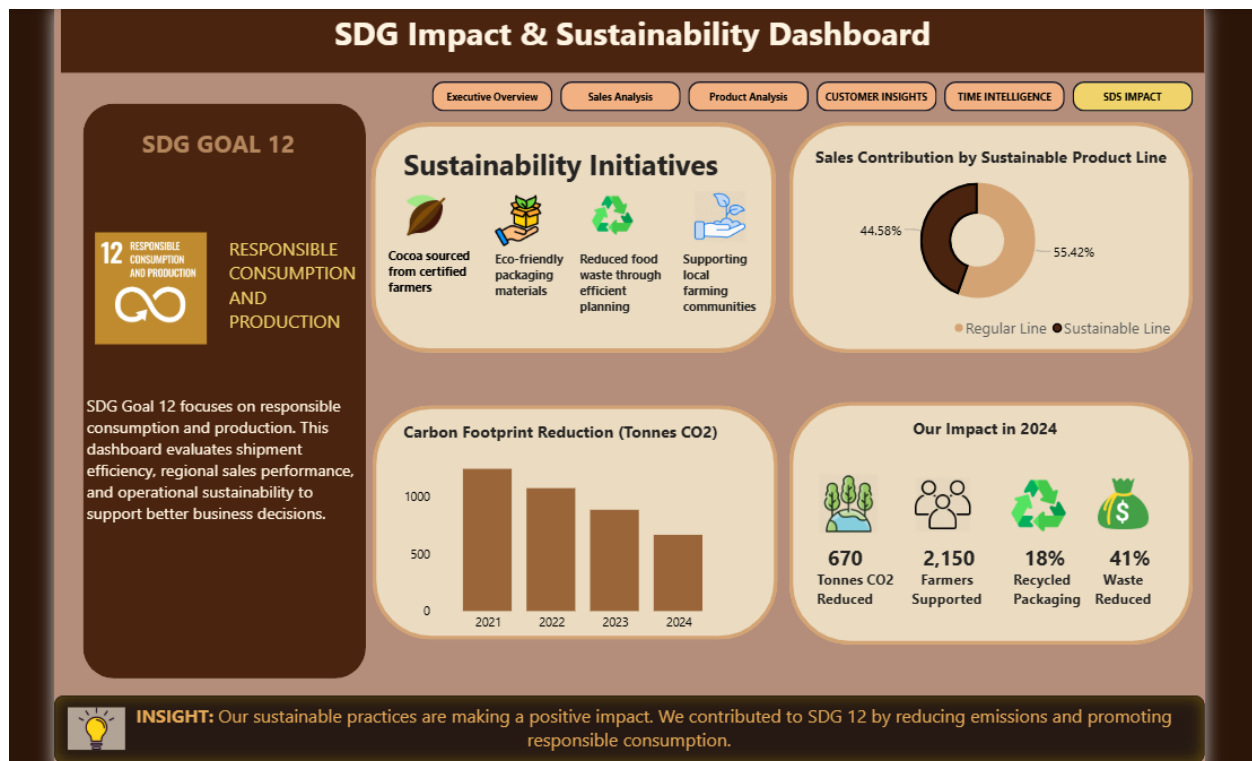
12. SDG MAPPING

Sustainable Development Goal Used

SDG Goal 12 – Responsible Consumption and Production

The project supports sustainability analysis by:

- Monitoring operational efficiency
- Reducing shipment waste
- Encouraging sustainable production
- Tracking environmental contribution metrics



13. KEY INSIGHTS

- Bars category contributes the highest sales revenue.
- Premium products generate strong profit margins.
- The Americas region shows the highest customer contribution.
- Monthly sales trends indicate stable business growth.
- Sustainability initiatives positively support operational efficiency.

14. CHALLENGES FACED

- Managing bookmark visibility
- Creating dynamic visual switching
- Designing consistent dashboard themes
- Handling DAX calculation errors
- Implementing time intelligence functions

15. LEARNINGS

Through this project, the following concepts were learned:

- Power BI Dashboard Development
- Data Modeling
- DAX Measures
- Time Intelligence Functions
- Dashboard Storytelling
- Interactive Report Design
- Bookmark and Navigation Implementation

16. CONCLUSION

The Chocolate Sales Analytics Dashboard successfully demonstrates the use of Power BI for business intelligence and data visualization. The project combines advanced DAX calculations, interactive dashboards, time intelligence analysis, bookmarks, and sustainability insights to create a professional analytics solution.

The dashboard transforms raw business data into meaningful visual insights that support effective decision-making and business storytelling.

REFERENCES

- Microsoft Power BI Documentation
- Kaggle Dataset Sources
- Power BI Community Forums
- Data Visualization Course Material

THANK YOU